

Endomembranes - Factoid Sheet

Week 4 - Wednesday - Core Lecture 9 - current faculty deck + official Week 4 objectives

Master chain: signal sequence → ER import/folding → COPII to Golgi → sorting → Rab/SNARE fusion → secretion, lysosome, or membrane.

ER Import

Hydrophobic signal → **SRP** binds ribosome and pauses translation → SRP receptor docks at ER → translocon opens → translation resumes into ER. Stop-transfer sequences anchor membrane proteins.

ER Quality Control

N-linked glycosylation occurs in ER lumen; chaperones monitor folding.

Persistent misfolding → retrotranslocation → ubiquitin → proteasome. Excess unfolded protein → **UPR**.

Coats

Coat	Route
COPII	ER → Golgi
COPI	Golgi → ER / retrieval
Clathrin	Endocytosis + Golgi/endosomal sorting

Fusion

Rab + Rab effector = docking specificity. **v-SNARE** + **t-SNARE** zipper together = fusion. **NSF** + **ATP** separates SNAREs afterward.

Scission

Dynamin-GTP pinches off clathrin-coated vesicles; BAR-domain proteins help membrane curvature.

Endomembranes - Fast Recall

Lysosome Targeting

GlcNAc phosphotransferase creates **mannose-6-phosphate** tag → M6P receptor in trans-Golgi → late endosome/lysosome.

Disease	Key defect
Hurler	Specific lysosomal enzyme missing
I-cell	GlcNAc phosphotransferase defect → broad M6P sorting failure
CF, F508del	CFTR misfolds in ER → degradation before surface trafficking
Familial hypercholesterolemia	Defective LDL uptake/receptor pathway

Secretion

Constitutive: default continuous exocytosis.

Regulated: cargo stored until stimulus (e.g., neurotransmitter, mast-cell granule).

Endocytosis

pH decreases PM → early endosome → late endosome → lysosome. LDL receptor unloads cargo then recycles; EGFR is commonly degraded to terminate signaling.

Large-Cargo Routes

Phagocytosis: macrophages/neutrophils ingest large particles, often after opsonization.

Autophagy: intracellular material ultimately delivered to lysosomes.

Must-Not-Miss

COPII OUT of ER; COPI back. Rab docks; SNARE fuses. M6P = lysosome. Dynamin = clathrin-pit scission. F508del = ER folding/trafficking defect.

Assigned ScholarRx

Endoplasmic Reticulum & Golgi Apparatus: <https://exchange.scholarrx.com/brick/endoplasmic-reticulum-and-golgi-apparatus>